

The Executive's 12-Question Materials Failure & Manufacturing Cost Checklist

A short guide for deciding whether a product failure, quality problem, material change or process issue requires independent technical review — or whether your own team should keep it.

This is not a sales document. Several of the twelve questions are designed to establish that a problem does *not* need outside help, which is the right answer more often than it is the profitable one. Work through it with the people who actually see the parts.

One instruction before you start: if failed parts still exist, stop anyone from cleaning, cutting, re-drying or shipping them back until you have finished Part 2. That single step preserves more investigative options than everything else in this document combined.

Part 1 — What is it actually costing?

Almost every one of these problems is escalated on a number that turns out to be a fraction of the real one. Establish the real one first, because it sets how much investigation is proportionate.

1. What is the fully loaded cost per month?

Scrap and rework are the visible part. Add returns and warranty reserve, expedited freight, held or quarantined inventory, line downtime, the engineering and quality hours already spent, and any price concession made to keep a customer. Write down one monthly figure.

Why it matters: If you cannot produce that figure in an afternoon, that is itself a finding: nobody currently owns the problem end to end.

2. What decision is waiting on the answer?

Releasing a held batch. Approving or reversing a material substitution. Requalifying a supplier. Committing tooling. Settling or contesting a claim. Name the decision explicitly.

Why it matters: "Why did it fail" has answers at many depths. The decision is what determines how deep the investigation has to go — and therefore what it should cost.

3. How long has this been open, and what has already been spent on it?

Count the engineering hours, the trial runs, the sample shipments and the meetings. Compare that with the cost of a scoped external investigation.

Why it matters: A problem that has consumed six months of internal attention without converging is usually not going to converge with more of the same.

Part 2 — What evidence still exists?

This is the time-critical section. Evidence is lost through ordinary, well-intentioned handling far more often than through neglect, and once it is gone no amount of budget recovers it.

4. Have the failed parts been preserved exactly as found?

Not cleaned, not cut, not re-dried, not re-processed, not returned to the supplier. Fracture surfaces in particular carry the signature of the failure mechanism and are destroyed by handling.

Why it matters: If parts have already been altered, record what was done and by whom. That record is itself evidence, and concealing it is far worse than the alteration.

5. Do you have retains from the material lot in question?

Unused material from the same lot, with its certificate of analysis, packaging and labels. Also retains from a lot that was working, if one exists — a comparison is worth more than an absolute measurement.

Why it matters: Without a retain, a material-related cause can often be inferred but not demonstrated. With one, it can usually be settled.

6. Do you have the process data from the period in question?

Machine settings, drying records, melt and mould temperatures, cycle times, residence time, regrind fraction, tool and cavity identity, shift and operator, and the maintenance log.

Why it matters: Data that exists only in an operator's memory is not data. Extract and date-stamp it now rather than after someone leaves.

Part 3 — What do you actually know about the cause?

The aim here is to separate what has been demonstrated from what has been asserted confidently for long enough that it now sounds demonstrated.

7. Is this one failure mode, or several?

Sort the rejects by defect type rather than counting them. Then sort by machine, tool, cavity, shift and material lot.

Why it matters: A single mechanism and three unrelated ones look identical on a scrap report and need completely different responses. This breakdown is frequently the whole diagnosis.

8. What changed before it started?

Resin lot or grade, supplier, additive or masterbatch package, regrind practice, tooling or a tool repair, drying equipment, ambient humidity or season, an operator, a shift pattern, a cleaning agent, a downstream customer's use.

Why it matters: The date the problem started is usually the single most informative fact available, and it is usually knowable to within a week.

9. Has any explanation been tested, or only argued?

For each hypothesis on the table, write down what evidence supports it and what evidence would refute it.

Why it matters: A hypothesis that nothing could refute is not a hypothesis. If every candidate explanation on your list is in that category, the investigation has not started yet.

Part 4 — What is the exposure?

The questions people leave until last, and the ones that most often determine whether the investigation should have been scoped differently from the beginning.

10. Is anyone outside the company already asking about cause?

A customer, an insurer, a regulator, a distributor, or counsel for any of them. Note who, when they first asked, and what has been said to them in writing.

Why it matters: Informal written reassurance given early is one of the most common ways a manageable quality problem becomes an expensive legal one.

11. Could this become a product-liability, IP or supply dispute?

Consider injury or property damage, a contract with a specification or conformance clause, a competitor's patent, or a supplier who has already denied responsibility.

Why it matters: If the answer is anything other than a confident no, the investigation should be scoped for that possibility now. Work commissioned as a plant investigation and repurposed as evidence later is materially weaker.

12. If your current best explanation is wrong, what does that cost?

Price the consequence of acting on the current theory and being mistaken: an ineffective corrective action, a recall that did not need to happen, a supplier wrongly blamed, or a claim conceded that could have been defended.

Why it matters: This is the number that decides whether independent review is worth commissioning. If it is small, keep the problem in-house.

How to read your answers

There is no score. What matters is the pattern:

Mostly confident answers, exposure low. Keep it in-house. You have the evidence, the history and the cost picture, and the consequence of being wrong is affordable. An outside view would add cost without adding much.

Confident on cost, thin on evidence. Act on Part 2 today. Preserving parts, retains and process data costs almost nothing this week and cannot be done at all next month.

Several competing explanations, none tested. This is the case where independent review usually pays for itself, because the value is in discriminating between hypotheses rather than in generating more of them.

Any yes in Part 4. Scope the work for that possibility from the start, and take advice before putting further conclusions in writing.

The next 48 hours

Whatever else you decide: quarantine and photograph the failed parts as found, secure a retain from the material lot involved, export the process data for the period, and write down the date the problem started and what changed around it. None of that commits you to anything, and all of it becomes impossible later.

If the pattern points to independent review:

Joshua U. Otaigbe, PhD, CEng, FIMMM

Founder & Principal, Flaney Associates, LLC

Manufacturers and product teams: info@flaneyassociates.com | +1 (601) 402-7282

Attorneys (confidential conflict check): jotaigbe@flaneyassociates.com | +1 (601) 451-8452

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